Project Design Phase

Solution Architecture

Date	23 March 2026
Team ID	69c103091466f700a6d324ef
Project Name	Smart Library Request Workflow in ServiceNow
Maximum Marks	4 Marks

Solution Architecture Overview

The **Smart Library Request Workflow** is designed to automate library operations, improve efficiency, and provide real-time transparency for both students and library staff. The solution bridges the gap between **manual library processes** and **modern digital workflow automation** using ServiceNow as the primary platform.

Goals of the Solution Architecture:

- Identify the **optimal technological solution** for library request management.
- Describe the **system structure, components, and data flow** for stakeholders.
- Define **features, modules, and solution requirements** for development.
- Provide **technical specifications** for implementation, scalability, and integration.

Architecture Components & Data Flow

1. **User Layer**
    - Users (students/faculty) interact with the system via **web portal or mobile browser**.
    - Users submit requests, track status, and receive notifications.
2. **Application / Workflow Layer**
    - **ServiceNow Flow Designer** manages request workflows, approvals, and notifications.
    - **Business Rules & Scripted Actions** automate decision-making, approvals, and alerts.
3. **Integration Layer**

- **External APIs** handle authentication (Gmail OAuth, LinkedIn OAuth).
- Email notifications and push notifications are integrated via ServiceNow Notification Engine.

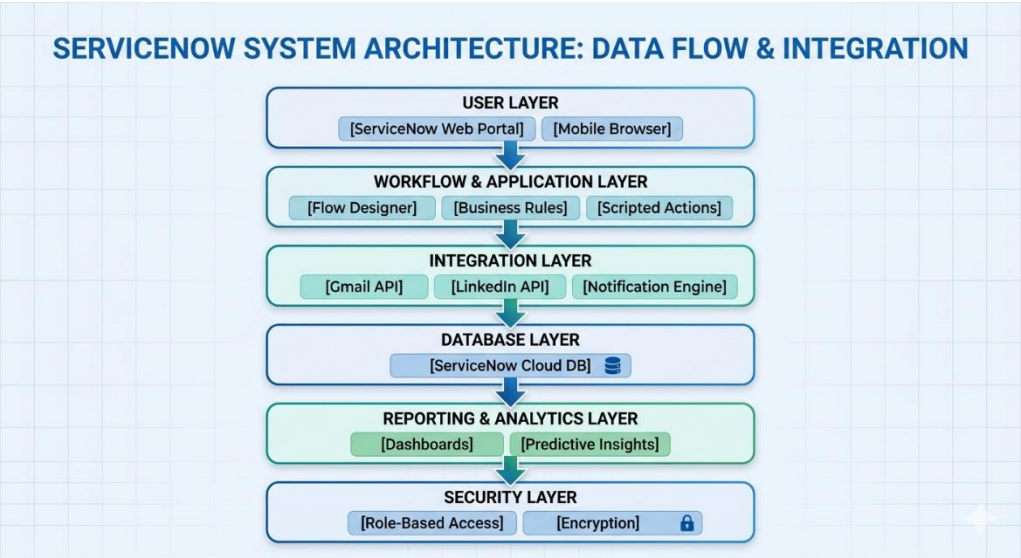
4. **Database Layer**
   - **ServiceNow Cloud Database** stores user profiles, request data, book inventory, and logs.
   - Ensures **data consistency, security, and scalability**.
5. **Reporting & Analytics Layer**
   - Admin dashboards provide insights on request trends, overdue books, and usage patterns.
   - Optional **predictive recommendations** for book suggestions based on user history.
6. **Security Layer**
   - Role-based access control (RBAC) ensures students and librarians access only relevant data.
   - Data encrypted at rest and in transit (HTTPS, SHA-256).

Data Flow Description

Step-by-Step Flow:

1. User logs in via ServiceNow portal (or Gmail/LinkedIn OAuth).
2. User submits a book request form → request enters **workflow engine**.
3. Workflow engine triggers **approval process** for librarian.
4. Librarian approves/rejects → system sends **automated notification** to user.
5. Approved requests update **database and reporting dashboards**.
6. Reminders sent for overdue books → analytics report updated automatically.

Proposed Architecture Diagram



Reference Example: AWS Voice Patient Diary Architecture